

TASK 2 - FUZZY INFERENCE SYSTEM

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Fuzzy Inference System for Production Planning

Case study: Daily production of ABC canned food | Tsukamoto Method

1. Problem Statement

A canned food company produces type ABC food. Based on last month's data, the daily **demand** ranged between 1,000 and 5,000 packages/day, and the daily **inventory** in the warehouse ranged between 100 and 600 packages/day. Due to operational constraints, the company can produce a maximum of 7,000 packages/day and, for machine and workforce efficiency, must produce at least 2,000 packages/day. The production decision is governed by four fuzzy rules:

Rule	IF (Demand)	AND (Inventory)	THEN (Production)
R1	DECREASES	MANY	DECREASES
R2	DECREASES	FEW	DECREASES
R3	INCREASES	MANY	INCREASES
R4	INCREASES	FEW	INCREASES

Question: how many packages of ABC food must be produced if the current demand is **4,000** packages and the current warehouse inventory is **300** packages?

Because both linguistic terms of the consequent variable (Production DECREASES / INCREASES) are monotonic, this problem is solved with the **Tsukamoto fuzzy inference method**: each rule produces one crisp value from its own (monotonic) output set, and the final result is the firing-strength-weighted average of all rule outputs.

2. Fuzzy Variables and Membership Functions

2.1 Demand (domain: 1,000 – 5,000 packages/day)

Linear membership functions are defined for the two linguistic terms DECREASES and INCREASES:

$$\mu_{\text{DECREASES}}(x) = 1, \quad x \leq 1000$$

$$\mu_{\text{DECREASES}}(x) = (5000 - x) / 4000, \quad 1000 \leq x \leq 5000$$

$$\mu_{\text{DECREASES}}(x) = 0, \quad x \geq 5000$$

$$\mu_{\text{INCREASES}}(x) = 1 - \mu_{\text{DECREASES}}(x)$$

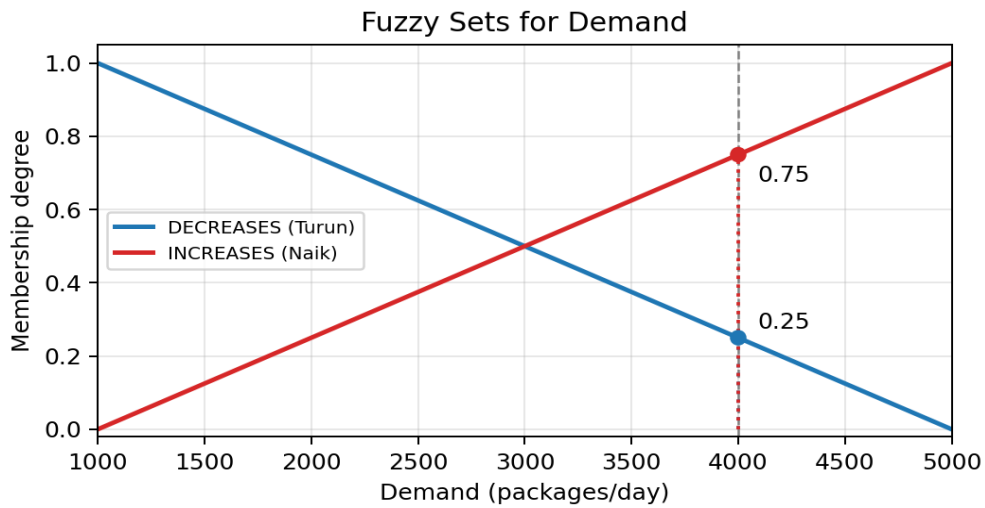


Figure 1. Fuzzy sets for Demand, with the crisp input $x = 4000$ marked.

2.2 Inventory (domain: 100 – 600 packages/day)

$$\mu_{\text{FEW}}(y) = 1, \quad y \leq 100$$

$$\mu_{\text{FEW}}(y) = (600 - y) / 500, \quad 100 \leq y \leq 600$$

$$\mu_{\text{FEW}}(y) = 0, \quad y \geq 600$$

$$\mu_{\text{MANY}}(y) = 1 - \mu_{\text{FEW}}(y)$$

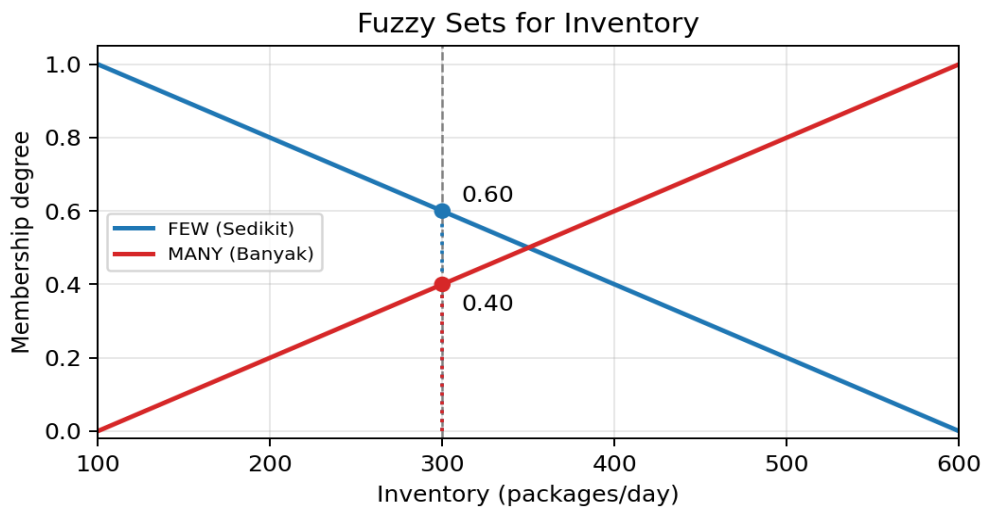


Figure 2. Fuzzy sets for Inventory, with the crisp input $y = 300$ marked.

2.3 Production (domain: 2,000 – 7,000 packages/day)

$$\mu_{\text{DECREASES}}(z) = 1, \quad z \leq 2000$$

$$\mu_{\text{DECREASES}}(z) = (7000 - z) / 5000, \quad 2000 \leq z \leq 7000$$

$$\mu_{\text{DECREASES}}(z) = 0, \quad z \geq 7000$$

$$\mu_{\text{INCREASES}}(z) = 1 - \mu_{\text{DECREASES}}(z)$$

Since these functions are monotonic and invertible, given a firing strength α the corresponding crisp production value can be solved directly:

$$z = 7000 - \alpha \times 5000 \quad (\text{if consequent is DECREASES})$$

$$z = 2000 + \alpha \times 5000 \quad (\text{if consequent is INCREASES})$$

3. Step 1 – Fuzzification of the Inputs

Given Demand = 4000 and Inventory = 300:

Variable	Formula	Result
$\mu_{\text{DEMAND-DECREASES}}(4000)$	$(5000 - 4000) / 4000$	0.25
$\mu_{\text{DEMAND-INCREASES}}(4000)$	$(4000 - 1000) / 4000$	0.75
$\mu_{\text{INVENTORY-FEW}}(300)$	$(600 - 300) / 500$	0.60
$\mu_{\text{INVENTORY-MANY}}(300)$	$(300 - 100) / 500$	0.40

4. Step 2 – Rule Evaluation (Inference)

Each rule combines its two antecedents with the fuzzy AND operator, implemented as the **minimum** of the membership degrees. This gives the firing strength α_i of rule i:

Rule	Antecedent computation	α_i
R1	$\min(\mu_{\text{DEC}}(4000), \mu_{\text{MANY}}(300)) = \min(0.25, 0.40)$	0.25
R2	$\min(\mu_{\text{DEC}}(4000), \mu_{\text{FEW}}(300)) = \min(0.25, 0.60)$	0.25
R3	$\min(\mu_{\text{INC}}(4000), \mu_{\text{MANY}}(300)) = \min(0.75, 0.40)$	0.40
R4	$\min(\mu_{\text{INC}}(4000), \mu_{\text{FEW}}(300)) = \min(0.75, 0.60)$	0.60

5. Step 3 – Per-Rule Crisp Output (Tsukamoto)

Each rule's consequent set is monotonic, so its own crisp value z_i is obtained by inverting the corresponding Production membership function at α_i :

Rule	Consequent	Computation	z_i (packages/day)
R1	DECREASES	$z1 = 7000 - 0.25 \times 5000$	5750.00
R2	DECREASES	$z2 = 7000 - 0.25 \times 5000$	5750.00
R3	INCREASES	$z3 = 2000 + 0.40 \times 5000$	4000.00
R4	INCREASES	$z4 = 2000 + 0.60 \times 5000$	5000.00

6. Step 4 – Defuzzification (Weighted Average)

The final crisp production quantity Z is the firing-strength-weighted average of the four per-rule outputs:

$$\begin{aligned}
 Z &= (\alpha_1 z_1 + \alpha_2 z_2 + \alpha_3 z_3 + \alpha_4 z_4) / (\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4) \\
 Z &= (0.25 \times 5750 + 0.25 \times 5750 + 0.40 \times 4000 + 0.60 \times 5000) / (0.25 + 0.25 + 0.40 + 0.60) \\
 Z &= (1437.5 + 1437.5 + 1600 + 3000) / 1.5 \\
 Z &= 7475 / 1.5 = \mathbf{4983.33}
 \end{aligned}$$

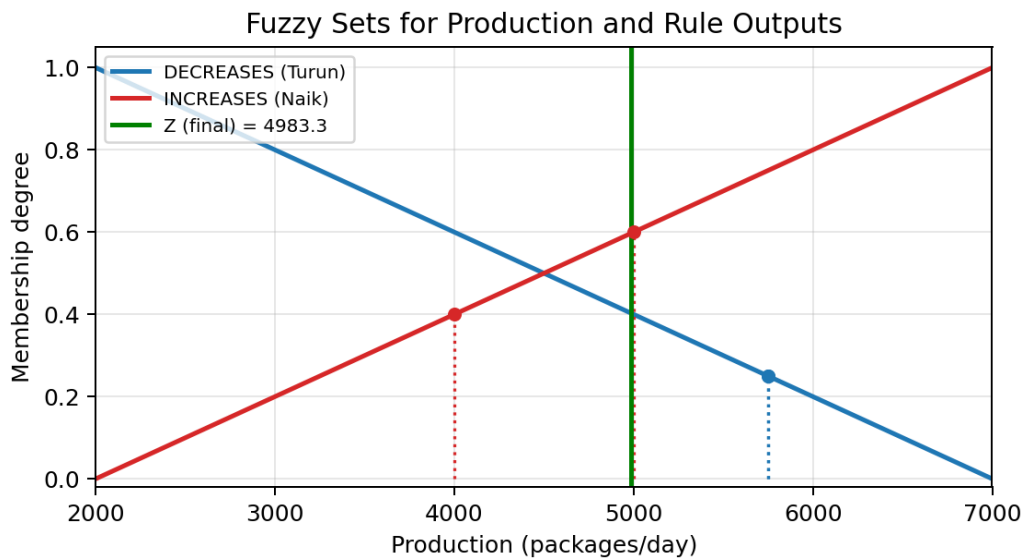


Figure 3. Production fuzzy sets with each rule's crisp output z_i at height α_i , and the final defuzzified value Z (green line).

7. Result and Interpretation

With Demand = 4,000 packages and Inventory = 300 packages, the fuzzy inference system recommends a production quantity of approximately
 $Z \approx 4,983.33 \approx 4,983$ packages/day.

This value lies comfortably within the company's feasible production range of [2,000, 7,000] packages/day, so the recommendation is operationally valid. Intuitively, the result makes sense: demand (4,000) is above the midpoint of its range and leans more toward “INCREASES” ($\mu = 0.75$) than “DECREASES” ($\mu = 0.25$), while inventory (300) is below its midpoint and leans more toward “FEW” ($\mu = 0.60$) than “MANY” ($\mu = 0.40$). Both leanings push toward the rules that increase production (R3, R4), and rule R4 (demand rising, inventory low) has the highest firing strength (0.60) of all four rules. As a result, the weighted average is pulled well above the midpoint of the production range (4,500), landing at 4,983 packages — close to, but appropriately short of, the company's maximum capacity.

8. Appendix – Verification Code

The manual computation above was cross-checked with the following Python script, which reproduces every intermediate value exactly.

```
import numpy as np

D_MIN, D_MAX = 1000, 5000    # Demand domain
I_MIN, I_MAX = 100, 600      # Inventory domain
P_MIN, P_MAX = 2000, 7000    # Production domain
demand0, inv0 = 4000, 300    # Given crisp inputs

def mu_demand_turun(x): return np.clip((D_MAX - x) / (D_MAX - D_MIN), 0, 1)
def mu_demand_naik(x): return 1 - mu_demand_turun(x)
def mu_inv_sedikit(y): return np.clip((I_MAX - y) / (I_MAX - I_MIN), 0, 1)
def mu_inv_banyak(y): return 1 - mu_inv_sedikit(y)
def z_turun(a): return P_MAX - a * (P_MAX - P_MIN)
def z_naik(a): return P_MIN + a * (P_MAX - P_MIN)

mu_Dt, mu_Dn = mu_demand_turun(demand0), mu_demand_naik(demand0)
mu_Is, mu_Ib = mu_inv_sedikit(inv0), mu_inv_banyak(inv0)

a1 = min(mu_Dt, mu_Ib)    # R1: Demand DECREASES & Inventory MANY -> Production DECREASES
a2 = min(mu_Dt, mu_Is)    # R2: Demand DECREASES & Inventory FEW -> Production DECREASES
a3 = min(mu_Dn, mu_Ib)    # R3: Demand INCREASES & Inventory MANY -> Production INCREASES
a4 = min(mu_Dn, mu_Is)    # R4: Demand INCREASES & Inventory FEW -> Production INCREASES

z1, z2 = z_turun(a1), z_turun(a2)
z3, z4 = z_naik(a3), z_naik(a4)

Z = (a1*z1 + a2*z2 + a3*z3 + a4*z4) / (a1 + a2 + a3 + a4)
print(f"alphas = {a1:.2f}, {a2:.2f}, {a3:.2f}, {a4:.2f}")
print(f"z's = {z1:.2f}, {z2:.2f}, {z3:.2f}, {z4:.2f}")
print(f"Z (recommended production) = {Z:.2f}" # -> 4983.33)
```

Running this script reproduces every number in Sections 3–6 exactly, confirming the final recommended production quantity of **4,983 packages/day**.